Spatial Resolution and Algorithmic Redistricting: A Test of the Modifiable Areal Unit Problem

Giovanni Della-Libera

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Abstract

All algorithmic redistricting methods depend on a choice of atomic spatial units—typically census tracts, block groups, or blocks. This choice introduces the Modifiable Areal Unit Problem (MAUP): different spatial aggregations can yield different results even with identical data. We empirically test whether key findings from automated redistricting are sensitive to spatial resolution by comparing recursive bisection at three census geographic levels: tracts (∼85,000 units), block groups (∼240,000), and blocks (∼11 million). [Results TBD]. This represents the first systematic MAUP analysis in algorithmic redistricting and establishes the generalizability—or lack thereof—of computational approaches to district design.

1 Introduction

1.1 The MAUP Problem

The Modifiable Areal Unit Problem (MAUP), first systematically analyzed by **?**, describes a fundamental challenge in spatial analysis: results can vary depending on how space is divided into units, even when the underlying data remains constant. This problem has two components: the *scale effect* (aggregating from fine to coarse resolution changes values) and the *zoning effect* (different boundaries at the same scale produce different results).

1.2 Redistricting Context

Congressional redistricting in the United States relies on census geographic units as atomic building blocks. The Census Bureau provides a nested hierarchy of spatial units: blocks (∼100 residents each), block groups (∼1,200 residents), and tracts (∼4,000 residents). Nearly all algorithmic redistricting research uses census tracts as the default aggregation level—a choice motivated by computational tractability, historical precedent, and balance between granularity and stability. However, this choice remains an untested assumption.

1.3 Portfolio Findings

Recent work on automated redistricting using recursive bisection has produced several key empirical findings:

- A 42% minority population threshold determines majority-minority district formation success (?)
- Algorithmic approaches can produce 69 more majority-minority districts than enacted plans (?)
- Edge-weighted partitioning improves compactness by 56% (?)
- Recursive partitioning provides greater temporal stability than n-way methods (?)

All of these findings were derived using census tracts as spatial units.

1.4 The Critique

A natural question arises: are these findings artifacts of tract-level aggregation, or do they persist at finer (block groups, blocks) or coarser resolutions? This question has both methodological and substantive implications. Methodologically, it tests the robustness of computational findings to unit choice. Substantively, it matters for redistricting practice: if finer resolutions yield better outcomes (more majority-minority districts, more compact shapes), practitioners should use the finest feasible resolution.

1.5 This Paper

We conduct the first systematic empirical test of MAUP in algorithmic redistricting by rerunning the complete redistricting pipeline at three spatial resolutions: census tracts (baseline), block groups (3× finer), and census blocks (130× finer). We test whether key findings from prior work replicate across resolutions or vary systematically with spatial scale.

2 Spatial Resolution in Redistricting

2.1 Census Geographic Hierarchy

The U.S. Census Bureau organizes geographic data in a nested hierarchy:

- **Census blocks**: The finest census unit, typically bounded by streets, with ∼11 million blocks nationally and mean population ∼100 residents
- **Block groups**: Aggregations of blocks, with ∼240,000 nationally and mean population ∼1,200 residents
- **Census tracts**: Aggregations of block groups, with ∼85,000 nationally and mean population ∼4,000 residents

Tracts are the standard unit for redistricting research due to historical precedent (**?**), computational tractability, and balance between spatial detail and statistical stability. However, state redistricting commissions often use blocks for official plans, making block-level analysis policy-relevant.

2.2 MAUP Theory

MAUP manifests through two mechanisms in redistricting:

Scale effect. Finer resolutions permit more precise boundary placement. A minority community concentrated within a census tract becomes spatially identifiable only at block-level resolution. Conversely, coarser resolutions smooth over heterogeneity—a 60% minority tract may contain blocks ranging from 20% to 90% minority.

Zoning effect. Even at fixed resolution, different boundary configurations produce different results. However, we hold boundaries fixed (using official census geography), so we isolate scale effects.

2.3 Prior Work on MAUP in Redistricting

MAUP has been acknowledged in redistricting literature but rarely tested empirically:

- **?** demonstrated that partisan analysis depends on unit choice
- **?** showed that ecological inference suffers from aggregation bias
- Recent work on ensemble methods (**?**) uses blocks but does not compare to tract-level results

Gap: No prior work systematically tests whether algorithmic redistricting findings are sensitive to resolution choice. We address this gap by conducting identical analyses at three spatial scales.

3 Methodology

3.1 Data Preparation

3.1.1 Census Tracts (Baseline)

We use census tract shapefiles and demographics from the 2020 Census Redistricting Data (PL 94-171 Summary File), yielding approximately 85,000 tracts nationally. Adjacency graphs were constructed using Rook contiguity (shared edges, excluding corner-only contact).

3.1.2 Block Groups (Finer Resolution)

Block group shapefiles and demographics were obtained from the Census Bureau TIGER/Line files. With approximately 240,000 block groups nationally ($\sim 3\times$ more units than tracts), this resolution provides intermediate granularity. Adjacency graphs were constructed identically to the tract-level approach.

3.1.3 Census Blocks (Finest Resolution)

Census blocks represent the finest available resolution with approximately 11 million blocks nationally ($\sim 130\times$ more units than tracts). Block-level shapefiles were obtained from TIGER/Line files, with demographics aggregated from the PL 94-171 file. This resolution poses significant computational challenges due to graph size (estimated 35 million edges nationally).

3.2 Algorithm Configuration

To isolate resolution effects, we maintain identical parameters across all spatial scales:

- **Population balance**: $\pm 0.5\%$ deviation from ideal district size
- **Edge-weighting**: $\alpha = 5.0$ weight multiplier for minority-minority edges
- **Minority threshold**: $\tau = 0.40$ (units with $\geq 40\%$ minority population)
- **Partitioning method**: METIS 5.1.0 recursive bisection with multilevel coarsening

This ensures that any observed differences arise from spatial resolution rather than parameter variation.

3.3 Experimental Design

We conduct redistricting for all 50 states at three resolutions (150 total runs) using 2020 Census data. To validate computational feasibility, we first test a 10-state subset:

- **High minority**: Texas, California, Florida, Georgia, New York
- **Low minority**: Vermont, Maine, West Virginia, New Hampshire, Idaho

If subset results confirm METIS scalability at block-level resolution, we proceed to the full 50-state analysis.

3.4 Metrics

For each state and resolution, we compute:

- **Majority-minority (MM) district count**: Districts where minority population $> 50\%$
- **Polsby-Popper compactness**: $\frac{4\pi \cdot \text{Area}}{\text{Perimeter}^2}$ (higher is more compact)
- **Population deviation**: Verify $\pm 0.5\%$ constraint maintained
- **Runtime**: Computational performance and scalability analysis

4 Results

4.1 Threshold Analysis Across Resolutions

We test whether the 42% minority population threshold identified in prior work (**?**) persists across spatial resolutions.

4.2 VRA Compliance Comparison

We compare majority-minority district counts across resolutions to test whether the +69 district surplus finding is resolution-dependent.

4.3 Compactness Sensitivity

We examine whether compactness metrics vary systematically with spatial resolution or remain scale-invariant.

4.4 Computational Performance

We document computational feasibility limits for block-level redistricting.

5 Discussion

5.1 Interpretation of Findings

5.2 Theoretical Implications

MAUP Mechanisms.

Algorithm Behavior.

5.3 Policy Implications

For Redistricting Commissions.

For Courts.

5.4 Limitations

- **Computational**: Block-level analysis may face memory constraints for largest states
- **Data quality**: Blocks have small sample sizes, introducing demographic noise
- **Scope**: Only recursive bisection tested; other partitioners may behave differently
- **Census years**: Focus on 2020; temporal consistency untested

5.5 Future Work

Promising directions include:

- Testing MAUP with other partitioning algorithms (SCOTCH, spectral methods)
- Examining different adjacency structures (queen contiguity, distance-based)
- Multi-resolution ensemble approaches
- Extension to multi-census temporal analysis

6 Conclusion

We have conducted the first systematic empirical test of the Modifiable Areal Unit Problem in algorithmic redistricting by comparing recursive bisection across three spatial resolutions: census tracts, block groups, and census blocks. [Summary of key findings TBD].

Contribution. This work establishes the generalizability—or lack thereof—of automated redistricting findings to different spatial scales. [If robust: Our results validate census tracts as appropriate atomic units for redistricting research, enabling confident extrapolation of computational findings to policy contexts.] [If sensitive: Our results demonstrate that spatial resolution is a critical methodological choice with substantive implications for district design and VRA compliance analysis.]

Implications. For researchers, this work [establishes/challenges] the robustness of algorithmic redistricting methods. For practitioners, it provides [validation/guidance] on resolution choice in automated district design. For courts, it [confirms/complicates] the use of computational evidence in redistricting litigation.

Final statement. The choice of spatial resolution is not merely a technical detail—it is a methodological decision with consequences for democratic representation.